

SAHO – Product Requirements Document

Phase 2 – Functional Requirements & Feature Specifications

2.0 Product Overview

This phase defines the complete functional blueprint of SAHO. Every feature is designed around one principle: reduce cognitive load during substance-use crises while providing compassionate AI-driven support.

2.1 SAHO NOW

The primary entry point to the application. The landing screen presents a single calming action button labelled 'SAHO NOW'. Users immediately choose an emotional state or activate voice mode. Emotional States: • I'm Craving • I'm Panicking • I'm Lonely • I'm Feeling Sick • I Feel Like Giving Up • Just Speak (Voice) Acceptance Criteria: • One-tap access • Voice starts within 2 seconds • AI responds within 3 seconds • Zero typing required • Structured JSON responses only.

Voice AI

Speech is converted into text and sent to Gemini. The AI returns structured JSON containing: • Risk Level • Detected Emotion • Compassionate Message • Three Recommended Actions • Guided Breathing Flag The frontend must never parse natural language directly.

2.2 Guided Breathing Engine

Immediately after activation the application displays a breathing animation while AI processes the request. Requirements: • Continuous breathing loop • Pause while user speaks • Resume automatically • Reduced-motion accessibility support.

2.3 AI Recovery Companion

Gemini generates: 1. Reassurance 2. Short explanation of current emotional state 3. Three simple recovery actions Rules: • Maximum 80 words • No judgmental language • One sentence per instruction.

2.4 Circle of Safety

Users configure up to five trusted contacts during onboarding. Emergency messages are AI-generated after user confirmation and may include optional location sharing. Requirements: • Never notify automatically unless explicitly enabled • Clear confirmation before sending • Privacy-first implementation.

2.5 Caregiver Mode

Dedicated workflow for caregivers. Emergency Categories: • Overdose • Aggressive Behaviour • Withdrawal • Panic Attack • Craving Teleprompter Mode Requirements: • Large typography • One instruction per screen • Swipe navigation • High contrast • Text-to-speech support.

2.6 Trigger Vision

Camera-based AI analyses surroundings for relapse triggers including alcohol, syringes, smoking equipment and related objects. Output: • Potential trigger • Reason • Suggested grounding exercise Requirements: • Camera opens within 2 seconds • Analysis completes within 5 seconds • Images are never stored without explicit consent.

2.7 Substance Identifier

Caregivers photograph unknown medication or substances. AI returns: • Possible identification • Common use • Side effects • Overdose warning signs • Confidence level If confidence is low the system must advise users not to consume the substance and seek professional verification.

2.8 Recovery Timeline

Instead of focusing on streaks, SAHO celebrates healthy decisions. Examples: • Completed breathing exercise • Contacted trusted person • Read educational content • Avoided relapse The AI also generates compassionate daily reflections.

2.9 Recovery Replay

Every completed support session is summarized privately to reinforce successful coping strategies and personal growth.

2.10 Educational Hub

Short AI-generated educational cards covering: • Cravings • Withdrawal • Safe medication practices • Overdose awareness • Harm reduction Cards include optional voice narration.

2.11 Offline Safety

Critical features remain available offline: • Emergency guidance • Breathing exercises • Naloxone instructions • Emergency contacts • Offline safety resources

2.12 Future Roadmap

Future capabilities include wearable integration, therapist dashboards, anonymous peer support, medication reminders and recovery analytics.

2.13 Global Functional Requirements

Every feature must: • Complete within three taps where possible • Respond within three seconds • Handle offline scenarios gracefully • Protect user privacy • Never expose AI prompts • Log errors safely without storing sensitive conversations.

2.14 MVP Acceptance Criteria

The MVP is complete when: • SAHO NOW functions with voice and touch • Guided breathing works • Caregiver teleprompter is operational • Vision AI identifies triggers and substances • Recovery timeline and replay function correctly • Educational hub is available • Offline emergency resources work • Accessibility and performance targets are achieved.